Design Convergence in the Post-AI Era:

Quantifying Visual Similarity Across Fashion Houses

Final Project Presentation

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Problem Statement

Background

Enabled by the data from e-commerce and social media, AI tools are now trained to detect emerging styles, accelerating the sketching process and help with material experiment (Deldjoo et al., 2022; Huang et al., 2018)

Before

Mood board

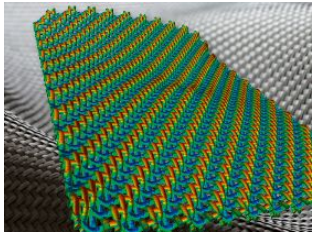
Sketches by Hand

Field Work: Pattern/Material Sampling

After

Trend Predicting Tool

AI-assisted sketch generation



Material/Pattern Simulation

Problem Statement

From the demand side: *73%* of the fashion executives stated that AI will be their priority in the future (BoF & McKinsey, 2024)

What we are seeing: Fashion as a human-centric, intuitive, and observational process (Gaimster, 2012) into a data-driven analytic capability with AI incorporated.

Tension Driven Motivation:

Beneficial for Innovation:

- ✓ Broaden the range of references designers can draw from;
- ✓ Freeing their mental capacity for more creative tasks;
- ✓ Increases the likelihood of generating unconventional combinations;

Harmful to Innovation:

- ✗ Algorithmic consensus may reduce diversity and encourage convergence;
- ✗ Weaken the tacit, embodied forms of knowledge central to creative discovery;

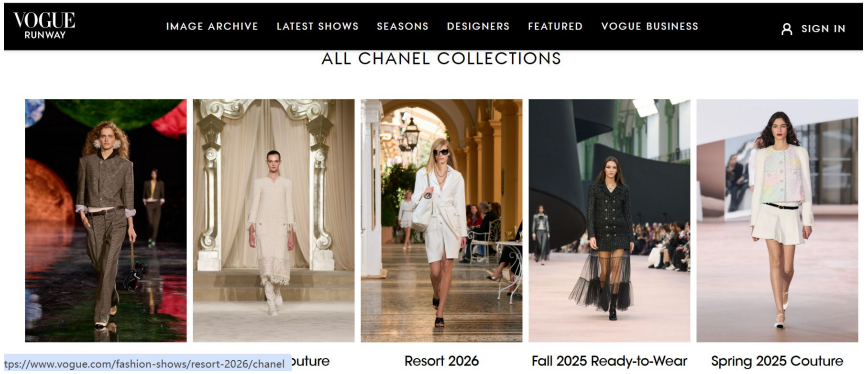
Hypothesis: AI tools adoption is accompanied by increased design similarity across different fashion houses as evidence of decreased novelty in the fashion industry.

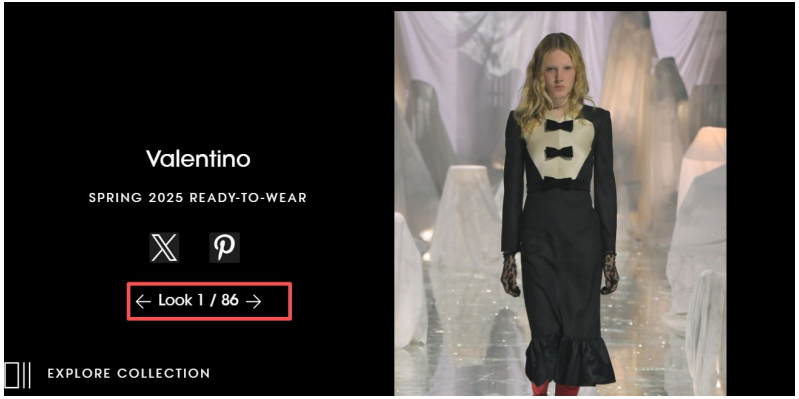
Data Collection

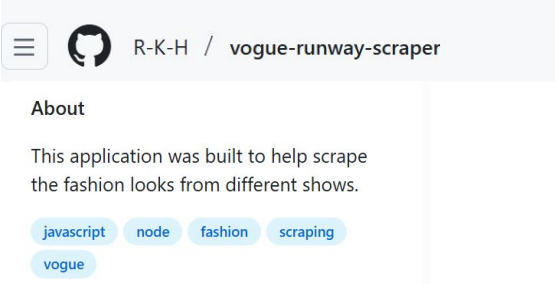
Datasource: *Vogue Runway*

- comprehensive public database of runway fashion shows, covering major fashion weeks and key designers from the 1990s to the present.

- Automate the image fetching process enabled by their standardized URLs and an open-source scraper





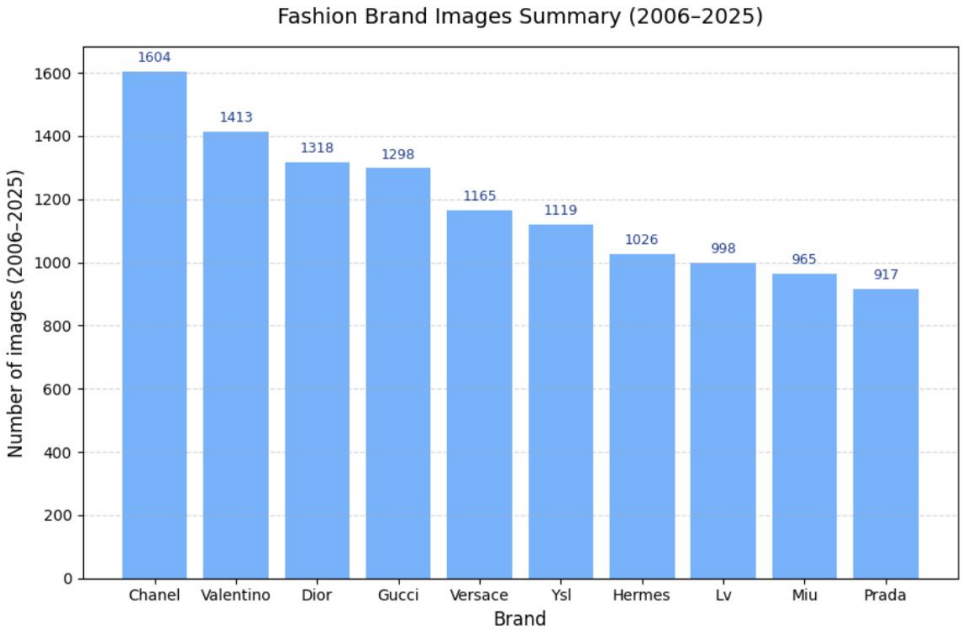


https://github.com/R-K-H/vogue-runway-scraper

Data Collection

Sample:

- randomly picked 10 comparable luxury clothing brands
- 20 spring ready-to-wear collections from 2006 to 2025 (Fall RTW; Spring RTW; Resort; Couture...)
- 45-80 images per brands per season

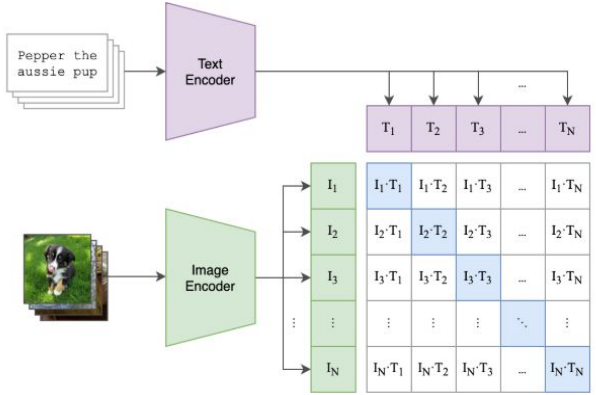


Data Cleaning

Fashion Clip (Chia et al., 2022): CLIP-style model fine tuned for fashion product hosted on Hugging Face

Contrastive Language - Image Pre-training
- OpenAI (2021): designed to learn a shared representation space for images and text.
- Trained on pairs of (image, text)

Fashion CLIP:
further **fine tuned** on 1.28 million *Farfetch* product image and description



Similarity score for each pair dot product (Ii, Ti)

Make the diagonal cells as larger as possible

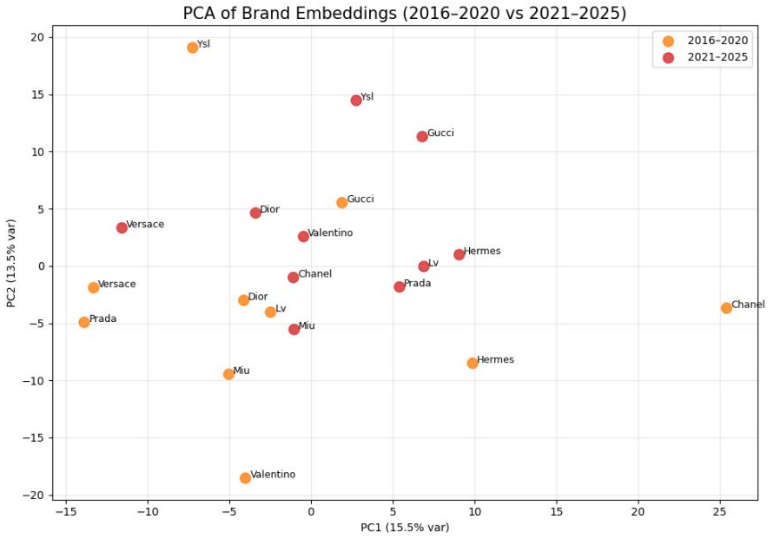
Off-diagonal as small as possible

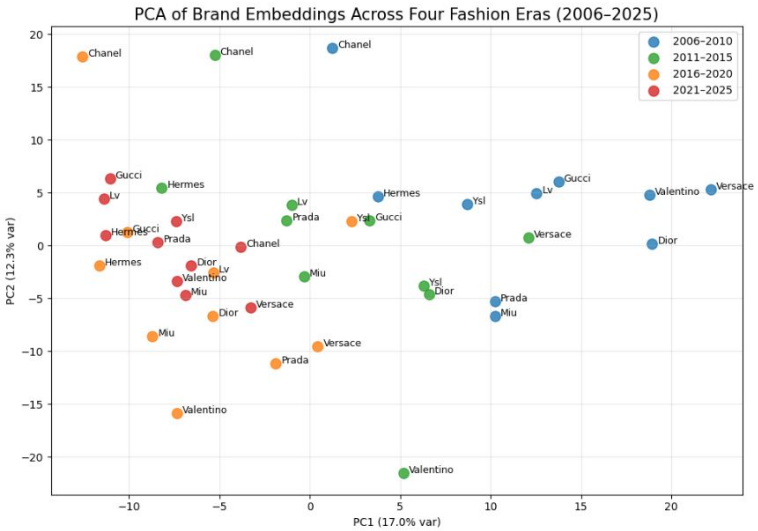
What's in the final dataset?
Each of the images is transformed into a **vector in a shared semantic space** where the distance among them are meaningful.

https://github.com/openai/CLIP
https://huggingface.co/patrickjohncyh/fashion-clip

Finding: PCA Visualize Similarity Across Brands

Model Setup: For each brand and each year, averaged all 512-dimensional embedding vectors to get year-brand vector; Pick 2021 as the cutoff year; standardize the dimensions





Result: There seems to a contraction of stylistic variance, indicating increasing convergence across fashion houses over the 20 years within a five year window.

Finding: Pairwise Cosine Similarity Across Brands

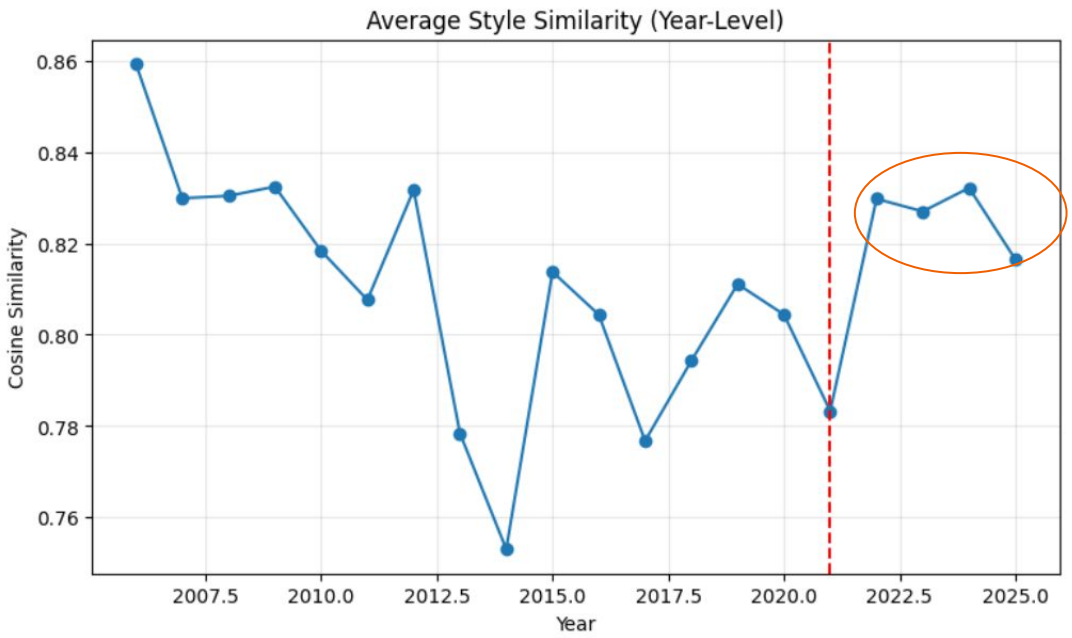
Pairwise Cosine Similarity:

Model setup:

- 10 brands yield 45 unique brand–brand similarity pairs.
- Average the 45 pairs of similarity within each year
- Plot the 20 years of data

Result:

Seems to have a jump after 2021



Finding: Pairwise Cosine Similarity Across Brands

Linear Regression:

Model Setup:

$$Similarity_t = \beta_0 + \beta_1 \cdot Trend_t + \beta_2 \cdot Post2021_t + \epsilon_t$$

Post2021 a dummy variable
Trend indexing each year

Result: there is a significant level increase in cross-brand similarity after 2021, net of any underlying temporal movement.

```
                            OLS Regression Results
==============================================================================
Dep. Variable:                avg_sim   R-squared:                       0.278
Model:                            OLS   Adj. R-squared:                  0.193
Method:                 Least Squares   F-statistic:                     3.269
Date:                Thu, 20 Nov 2025   Prob (F-statistic):             0.0629
Time:                        15:37:11   Log-Likelihood:                 49.218
No. Observations:                  20   AIC:                            -92.44
Df Residuals:                      17   BIC:                            -89.45
Df Model:                           2
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const          0.8356      0.012     69.582      0.000       0.810       0.861
post2021       0.0404      0.018      2.306      0.034       0.003       0.077
trend         -0.0032      0.001     -2.461      0.025      -0.006      -0.000
==============================================================================
Omnibus:                        7.269   Durbin-Watson:                   1.908
Prob(Omnibus):                  0.026   Jarque-Bera (JB):                5.064
Skew:                          -1.195   Prob(JB):                       0.0795
Kurtosis:                       3.604   Cond. No.                         45.5
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

Significant

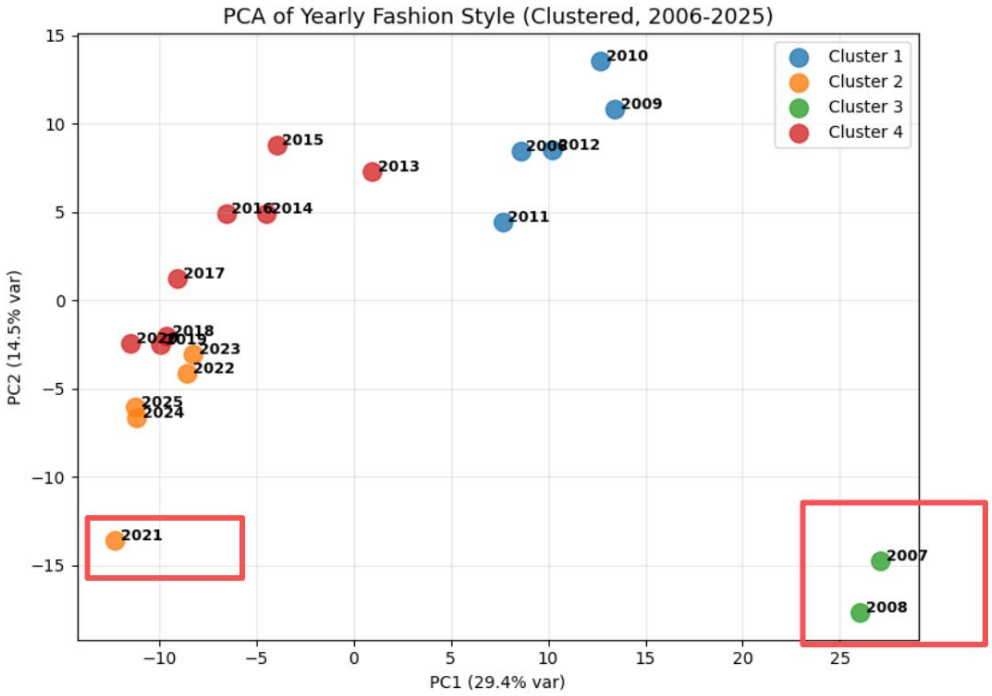
Additional Findings: Temporary Similarity

Temporary Similarity

If further aggregated all the brands for each year and apply PCA, we can show the temporary similarities. K-means clustering (k=4)

It seems that 2007, 2008 and 2021 are the years that stand out as major stylistic outliers relative to all other years.

Going through financial crisis, pandemic…



Conclusions & Limitations

Conclusions:

Consistent with the hypothesis that fashion houses in post AI-era is associated with higher visual similarity in their runway pieces.

Limitations: Toy project to practice the methods with MANY limitations

- The use of 2021 as a cutoff year is inherently arbitrary. There is no direct evidence on when, or even whether, each brand adopted AI-assisted design tools.
- Fashion-CLIP focuses more on the piece features but not able to capture the overall vibe, stage and other contextual settings.
- PCA visualization captures only 30% of the underlying variation in the embeddings
- For the regression analysis, the lack of control variables due to the feasibility of collecting additional data also limits validity.
- Small sample size

Appendix

References

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Vogue Runway. (n.d.). *Vogue Runway fashion show archive*. Condé Nast. https://www.vogue.com/fashion-shows

Appendix

Detailed Data & Method

S1: Use Github Scraper tool to fetch pictures from Vogue Runway (both cited)

- Clone the repository to my local device and run
- code will be submitted as file *fetch pics.ipynb*

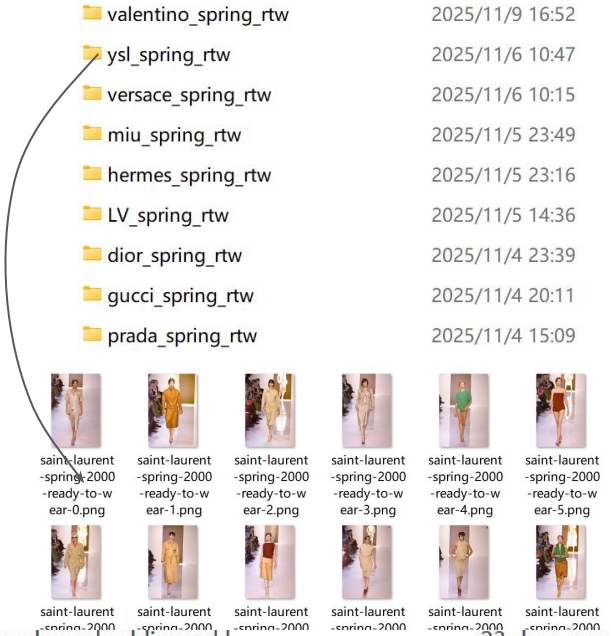
S2: Use Fashion-CLIP from Hugging Face (cited) to turn each picture into embedding

- Load the focal model and processor
- Pass each image through the model
- Store embeddings into a pandas Dataframe in a .pkl file
- Code will be submitted as file *pic_to_vector.ipynb*

S3: standardized all dimensions and PCA (n_componenets=2) and plot it

S4: Pairwise similarity & regression cosine_similarity (V1,V2) and put it with trend into a regression model

S5: temporary PCA at year level and clustering (k=4)

S3-S5 are saved in file *finalcode.ipynb*

Note: the raw data are images, so it is too big to submit in Canvas.